

Malithi Mithsara

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 [Malithi Mithsara](#) |  [Malithi-gif](#)

Carbondale, Illinois - 62901, USA

OBJECTIVE

Seeking a challenging role in data science to utilize my expertise in computer science, statistics, and machine learning. Passionate about applying advanced analytical techniques to solve real-world problems and contribute to innovative projects at the intersection of data-driven insights and intelligent systems.

EXPERIENCE

- Southern Illinois University.**  January 2024 - Current
Graduate Teaching Assistant Carbondale, USA
 - Teaching assistant for multiple computer science courses, including Operating Systems, Web Application Development, Machine Learning, and Soft Computing.
 - Coordinated ABET documentation by organizing, formatting, and maintaining course files (syllabi, assignments, assessments, and reports) to ensure compliance with accreditation requirements.
 - Researched large and small language models (LLMs/SLMs), utilizing fine-tuning and parameter-efficient techniques on multimodal data (text, vision, sensor) to address diverse real-world problems.
- Informatics Institute of Technology.**  August 2023 - December 2023
Assistant Lecturer Colombo, Sri Lanka
 - Taught courses in object-oriented programming, fundamentals of computer systems, software development, and advanced mathematics for data science students in ICT and computer science.
 - Supervised final-year research projects, guiding them from proposal to completion.
- Rajarata University of Sri Lanka.**  August 2021 - August 2023
Temporary Lecturer Mihintale, Sri Lanka
 - Taught Web Development, Operating Systems, and Management Information Systems to second-year ICT and Computer Science students.
 - Supervised and co-supervised final-year projects for third- and fourth-year students, guiding them from proposal to completion.
- Open University of Sri Lanka.**  October 2021 - August 2023
Online Tutor Colombo, Sri Lanka
 - Taught courses in web development, computer security, Python programming, and mobile application development to undergraduate students.
- Sri Lankan Institute of Information Technology, Sri Lanka.**  October 2020 - July 2021
Research Assistant Malabe, Sri Lanka
 - Supervised undergraduate research students at SLIIT while conducting independent research in computer vision, machine learning, and graph neural networks.
- Department of Meteorology, Sri Lanka.**  March 2020 - October 2020
Information Technology Officer Colombo, Sri Lanka
 - Maintained and updated the Department of Meteorology's website.
- University of Sri Jayewardenepura, Sri Lanka.**  January 2020 - March 2020
Research Assistant Colombo, Sri Lanka
 - Developed research objectives, prepared project proposals, conducted both individual and collaborative research projects, and supervised undergraduate research students.
- University of Sri Jayewardenepura, Sri Lanka.**  June 2017 - December 2019
Teaching Assistant Colombo, Sri Lanka
 - Conducted regular tutorial and practical classes; supervised undergraduate students (1st–4th year) in Object-Oriented Programming (Java, C++, and C), Database Management Systems (DBMS), and Web Technologies.

EDUCATION

- Southern Illinois University** January 2024 - Present
PhD in Computer Science Carbondale, USA
 - GPA: 4.00/4.00
 - Advisor: Prof. Ning Yan, Co-Advisor: Prof. A. R. Shahid
- University of Sri Jayewardenepura** January 2018 - July 2020
MSc in Computer Science Nugegoda, Sri Lanka
 - Grade: 3.30/4.00
- University of Sri Jayewardenepura** March 2014 - November 2016
Bachelor of Science General degree in Physical Science Nugegoda, Sri Lanka
 - GPA: 2.90/4.00

RESEARCH AND INTERESTS

My research focuses on human activity recognition (HAR) through large language models (LLMs) and small language models (SLMs), with a special emphasis on security and privacy challenges in LLM-integrated sensing systems. I explore how LLMs can be utilized to interpret complex sensor data for real-time activity analysis, while also investigating their vulnerabilities to various attacks, such as data poisoning, adversarial examples, and prompt injection attacks. My work aims to develop a privacy-preserving HAR pipeline that combines accurate sensing with secure and trustworthy AI-driven decision-making, especially in sensitive domains like healthcare and smart environments.

PROJECTS

- **Project A: [Automated Fake News Detection]** March 2025 - May 2025
Tools: [Jupyter notebook, Python] [🔗]
 - Conducted comparative analysis of machine learning and deep learning models for fake news detection, including CNN, LSTM, DistilBERT, FineBERT, RoBERTa, ALBERT, ELECTRA, Gradient Boosting, and Random Forest.
- **Project B: [AI-Powered YouTube Surveillance and Crime Detection]** March 2024
Tools: [Jupyter notebook, Python] [🔗]
 - Developed Python-based code for detecting crime-related YouTube videos, utilizing the YouTube API to automatically download and preprocess video datasets for analysis and classification.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] Malithi Wanniarachchi Kankanamge, et al. (2025). **Breaking Sensor-Text Alignment: Cross-Modal Attack on Contrastive Multimodal Human Activity Recognition**. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR 2026)*.
- [C.1] Malithi Wanniarachchi Kankanamge, et al. (2025). **Development of a Fall Detection and Safety Communication System Using Small Language Models**. In *Proceedings of 42nd International Symposium on Automation and Robotics in Construction (ISARC)*, July 28-29, Montreal, Canada. DOI: 10.22260/ISARC2025/0077.
- [C.1] WKM Mithsara, et al. (2024). **Zero-Shot Detection and Sanitization of Data Poisoning Attacks in Wearable AI Using Large Language Models**. In *the 23rd International Conference on Machine Learning and Applications (ICMLA)*, December 18-24, Miami, Florida, USA. DOI: 10.1109/ICMLA61862.2024.00233.
- [C.1] Malithi Wanniarachchi Kankanamge, et al. (2024). **Intelligent Fall Detection and Emergency Response for Smart Homes Using Language Models**. In *23rd International Conference on Machine Learning and Applications (ICMLA)*, December 18-24, Miami, Florida, USA. DOI: 10.1109/ICMLA61862.2024.00037.
- [C.1] WKM Mithsara, et al. (2024). **Large Language Model Integrated Healthcare Cyber-Physical Systems Architecture**. In *IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, July 02-04, Osaka, Japan. DOI: 10.1109/COMPSAC61105.2024.00228.
- [C.1] Malithi Wanniarachchi Kankanamge, et al. (2024). **Poster: Leveraging Large Language Models for Zero-Shot Detection and Mitigation of Data Poisoning in Wearable AI Systems**. In *the NeurIPS 2024 Workshop GenAI4Health*, December 12, Vancouver, Canada.
- [C.1] W.K.M Mithsara, et al. (2022). **Comparative Analysis of AI-Powered Approaches for Skeleton-Based Child and Adult Action Recognition in Multi-person Environment**. In *the International Conference on Computer Science and Software Engineering (CSASE)*, March 15 - 17.
- [C.1] W. K. Malithi Mithsara, et al. (2020). **A Sociable Robotic Platform to Make Career Advices for Undergraduates**. In *HCI International 2020 - Late Breaking Papers: Multimodality and Intelligence*, July 19–24, Copenhagen, Denmark.
- [J.1] W.K.M Mithsara, et al. (2020). **Blockchain-based Distributed Secure Crowdfunding and Decision-Making Platform for Large-scale Business Projects in Public and Private Sectors**. *European Modern Studies Journal*, Vol. 04, pp. 11.

SKILLS

- **Programming Languages:** JAVA, Python, C++, C, Language 5
- **Database Technologies:** MySQL, SQL (WAMP Server, XAMPP), Oracle
- **Web Technologies:** HTML, CSS, JavaScript, PHP, Web services (SOAP, REST), .NET
- **Mobile Development:** Android Application Development
- **Tools:** NetBeans, Eclipse, RapidMiner, Code Blocks, Maple, MATLAB, Minitab, Tableau, Unity
- **Research Skills:** Large Language Models and Small Language Models, Machine Learning and Deep Learning, Security and Privacy.

HONORS AND AWARDS

- **GPSC Travel and Conference Award**

Graduate and Professional Student Council

October 2024



- Through funding for conference travel, presentations, and professional development, it enables students to showcase their work, network, and gain visibility in their field.

- **BSN Travel Award**

IEEE-EMBS International Conference on Body Sensor Networks

October 2024



- Being selected in a competitive process backed by NSF and IEEE EMBS

LEADERSHIP EXPERIENCE

- **Secretary**

Sri Lankan Student Association (SLSA), SIUC

February 2025 - February 2026



- Prepare, circulate, and maintain meeting agendas and minutes.
- Coordinate international events for the society and the university.
- Maintain documentation (e.g., constitutions, membership lists).

- **Committee Member**

Statistics Society of the University of Sri Jayewardenepura, Sri Lanka.

January 2016 - November 2016



- Organizing events in collaboration with schools in Sri Lanka.
- Assisted in planning and coordinating workshops, seminars, and competitions.
- Supported fundraising activities and sponsorship coordination for society events.

PROFESSIONAL MEMBERSHIPS

- **Association for Computing Machinery:** Membership ID: 7351824

May 2024 - Present

- **IEEE:** Membership ID: 101772632

March 2026 - Present

- **Society of Women's Engineering, SIUC**

May 2025 - Present

- **Society of Women in Cybersecurity** Membership ID: 92375949

March 2026 - Present

ADDITIONAL INFORMATION

Languages: English (Intermediate level), **Interests:** Swimming, Badminton

REFERENCES

1. **Prof. Ning Yang,**

Professor in Information Technology, Faculty of Information Technology,
Southern Illinois University, Carbondale, USA.

Email: nyang@siu.edu

Relationship: [Advisor]

2. **Prof. A. R. Shahid**

Department of Computer Science,

School of Computing, Southern Illinois University, Carbondale, USA.

Email: abdurrahmanbin.shahid@siu.edu

Phone: +1-786-397-2558

Relationship: [Co-Advisor]

3. **Prof. R.G.N. Meegama,**

Senior Professor in Computer Science, Faculty of Applied Sciences,
University of Sri Jayewardenepura, Sri Lanka.

Email: rgn@sci.sjp.ac.lk

Phone: (+94) 11 275 8912

Relationship: [Teacher]